

FraudLens – GSE Risk Analysis Report

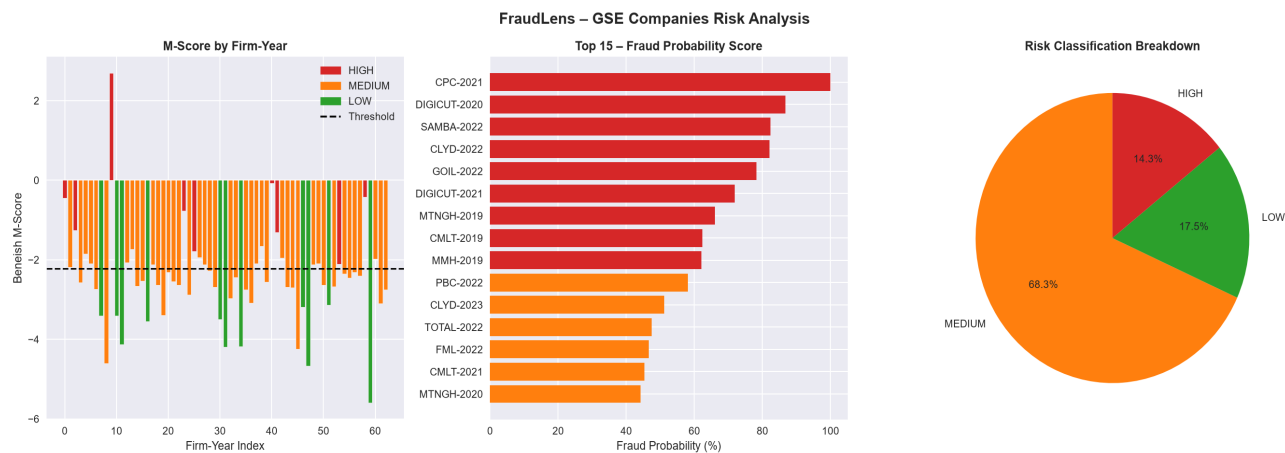
Ghana Stock Exchange Listed Companies | Analysis Date: 2026-10-10

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Executive Summary

FraudLens analysed 63 firm-year observations across 21 companies listed on the Ghana Stock Exchange covering the period 2019–2023. The pipeline applies the Beneish M-Score model, Z-score anomaly detection, and Isolation Forest machine learning to produce a composite fraud risk score for each firm-year. Of the 63 valid observations, 9 (14.3%) were classified HIGH risk, 43 (68.3%) MEDIUM risk, and 11 (17.5%) LOW risk.

Risk Visualisations



Full Risk Scoring Table

Firm-Year	Company	M-Score	B-Flag	Z-Flag	IF-Flag	Risk	Prob %
CPC-2021	Cocoa Processing Company	2.69	✓	✓	✓	HIGH	100.0
DIGICUT-2020	Digicut Advertising & Produc	-0.07	✓	✓	✓	HIGH	86.7
SAMBA-2022	Samba Foods Ltd	-0.43	✓	✓	✓	HIGH	82.4
CLYD-2022	Clydestone (Ghana) Ltd	-0.45	✓	✓	✓	HIGH	82.1
GOIL-2022	GOIL PLC	-0.77	✓	✓	✓	HIGH	78.3
DIGICUT-2021	Digicut Advertising & Produc	-1.30	✓	✓	✓	HIGH	71.8
MTNGH-2019	MTN Ghana (Scancom PLC)	-1.78	✓	✓	✓	HIGH	66.0
CMLT-2019	Camelot Ghana Ltd	-1.25	✓		✓	HIGH	62.4
MMH-2019	Meridian-Marshalls Holdings	-2.11	✓	✓	✓	HIGH	62.1
PBC-2022	Produce Buying Company Ltd	-2.44		✓	✓	MEDIUM	58.1
CLYD-2023	Clydestone (Ghana) Ltd	-2.19	✓	✓		MEDIUM	51.2
TOTAL-2022	TotalEnergies Marketing Ghan	-1.66	✓			MEDIUM	47.5
FML-2022	Fan Milk Ltd	-1.73	✓			MEDIUM	46.6
CMLT-2021	Camelot Ghana Ltd	-1.85	✓			MEDIUM	45.3
MTNGH-2020	MTN Ghana (Scancom PLC)	-1.93	✓			MEDIUM	44.2

DIGICUT-2022	Digicut Advertising & Produc	-1.96	✓			MEDIUM	43.9
DASPHARMA-2022	Dannex Ayrton Starwin Plc	-1.98	✓			MEDIUM	43.7
FML-2021	Fan Milk Ltd	-2.06	✓			MEDIUM	42.7
CMLT-2022	Camelot Ghana Ltd	-2.09	✓			MEDIUM	42.3
IIL-2020	Intravenous Infusions PLC	-2.09	✓			MEDIUM	42.3
TOTAL-2021	TotalEnergies Marketing Ghan	-2.10	✓			MEDIUM	42.2
IIL-2019	Intravenous Infusions PLC	-2.11	✓			MEDIUM	42.0
GGBL-2021	Guinness Ghana Breweries Plc	-2.12	✓			MEDIUM	41.9
MTNGH-2021	MTN Ghana (Scancom PLC)	-2.12	✓			MEDIUM	41.9
PBC-2021	Produce Buying Company Ltd	-2.97		✓		MEDIUM	41.7
MTNGH-2022	MTN Ghana (Scancom PLC)	-2.28				MEDIUM	40.1
SAMBA-2020	Samba Foods Ltd	-2.31				MEDIUM	39.7
GOIL-2019	GOIL PLC	-2.31				MEDIUM	39.6
MMH-2023	Meridian-Marshalls Holdings	-2.35				MEDIUM	39.2
SAMBA-2021	Samba Foods Ltd	-2.41				MEDIUM	38.5
SAMBA-2019	Samba Foods Ltd	-2.46				MEDIUM	37.9
GGBL-2019	Guinness Ghana Breweries Plc	-2.53				MEDIUM	37.0
GOIL-2020	GOIL PLC	-2.54				MEDIUM	36.9
TOTAL-2023	TotalEnergies Marketing Ghan	-2.56				MEDIUM	36.7
GGBL-2023	Guinness Ghana Breweries Plc	-3.40		✓		MEDIUM	36.6
CMLT-2020	Camelot Ghana Ltd	-2.57				MEDIUM	36.5
HORDS-2021	Hords Ltd	-4.24		✓	✓	MEDIUM	36.4
IIL-2021	Intravenous Infusions PLC	-2.63				MEDIUM	35.8
GOIL-2021	GOIL PLC	-2.63				MEDIUM	35.8
GGBL-2022	Guinness Ghana Breweries Plc	-2.63				MEDIUM	35.7
FML-2023	Fan Milk Ltd	-2.66				MEDIUM	35.4
IIL-2023	Intravenous Infusions PLC	-2.67				MEDIUM	35.3
HORDS-2019	Hords Ltd	-2.68				MEDIUM	35.2
MTNGH-2023	MTN Ghana (Scancom PLC)	-2.69				MEDIUM	35.1
HORDS-2020	Hords Ltd	-2.69				MEDIUM	35.0
CMLT-2023	Camelot Ghana Ltd	-2.74				MEDIUM	34.5
TOTAL-2019	TotalEnergies Marketing Ghan	-2.75				MEDIUM	34.4
DASPHARMA-2023	Dannex Ayrton Starwin Plc	-2.74				MEDIUM	34.4
GOIL-2023	GOIL PLC	-2.88				MEDIUM	32.7
CPC-2020	Cocoa Processing Company	-4.60		✓	✓	MEDIUM	32.0
TOTAL-2020	TotalEnergies Marketing Ghan	-3.08				MEDIUM	30.3
DASPHARMA-2023	Dannex Ayrton Starwin Plc	-3.09				MEDIUM	30.2
IIL-2022	Intravenous Infusions PLC	-3.13				LOW	29.8
HORDS-2022	Hords Ltd	-3.18				LOW	29.2

CPC-2019	Cocoa Processing Company	-3.41				LOW	26.4
CPC-2022	Cocoa Processing Company	-3.41				LOW	26.4
PBC-2019	Produce Buying Company Ltd	-3.49				LOW	25.4
GGBL-2020	Guinness Ghana Breweries Plc	-3.55				LOW	24.7
HORDS-2023	Hords Ltd	-4.67		✓		LOW	21.2
SAMBA-2023	Samba Foods Ltd	-5.60		✓	✓	LOW	20.0
CPC-2023	Cocoa Processing Company	-4.13				LOW	17.7
PBC-2023	Produce Buying Company Ltd	-4.17				LOW	17.2
PBC-2020	Produce Buying Company Ltd	-4.19				LOW	16.9

Key Findings

1. CPC-2021 (Cocoa Processing Company) recorded the highest fraud risk score of 100%, driven by an M-Score of +2.69 — far above the Beneish threshold — and confirmation from both anomaly detection methods. This firm-year warrants immediate further investigation.
2. Digidut Advertising & Production Ltd was flagged as HIGH risk in three consecutive years (2019–2021), suggesting persistent patterns of aggressive accounting rather than a one-off anomaly.
3. MTN Ghana (Scancom PLC) flagged HIGH risk in 2019 and 2020 during a period of rapid network expansion. This illustrates an important limitation of the M-Score: high capital expenditure and revenue growth in legitimate expansion phases can mimic fraud signals.
4. Produce Buying Company (PBC-2022) was cleared by the Beneish model but flagged by both Z-score and Isolation Forest, demonstrating the value of a multi-method approach over relying on a single model.

Disclaimer

This report is produced by FraudLens for academic and research purposes only. The classifications and fraud probability scores are generated by statistical models and do not constitute a finding of fraud, misconduct, or legal violation against any company or individual. Results should be interpreted alongside qualitative analysis and professional judgment. FraudLens is a portfolio project developed as part of an MSc in Forensic Accounting and Fraud Examination.